Documentation Packet Tracer

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Calculs du VLSM

Site utilisé : https://www.site24x7.com/tools/ipv4-subnetcalculator.html

Adresse IP utilisée : 192.168.99.0/24
Vlan 199 = min 50 Adresse IP
Vlan 299 = min 25 Adresse IP
Vlan 99 = min 10 Adresse IP

Vlan 199 /26 (max 62 host de 64 Adresse)
Adresse du réseau : 192.168.99.0/26
Host-Range : 192.168.99.1 - 192.168.99.62
Broadcast : 192.168.99.63

Vlan 299 /27 (max 30 host de 32 Adresse)
Adresse du réseau : 192.168.99.64/27
Host-Range : 192.168.99.65 - 192.168.99.94
Broadcast : 192.168.99.95

Vlan 99 /28 (max 14 host de 16 Adresse)
Adresse du réseau : 192.168.99.96/28
Host-Range : 192.168.99.97 - 192.168.99.110
Broadcast : 192.168.99.111

Configuration des Switches et Routeurs

- SW_Floor1

```
Current configuration : 1352 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
enable secret 5 $1$mERr$oF4ioZ7MAinG0VI8p.35b0
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport access vlan 10
!
interface FastEthernet0/2
 switchport access vlan 20
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
```

```
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
 no ip address
 shutdown
!
interface Vlan10
 no ip address
!
interface Vlan199
 ip address 192.168.99.3 255.255.255.192
!
ip default-gateway 192.168.99.1
!
!
!
!
line con 0
!
line vty 0 4
 password test..123
 login
line vty 5 15
 password test..123
 login
!
!
!
!
end
```

- SW_Floor2

```
Current configuration : 1318 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
enable secret 5 $1$mERr$oF4ioZ7MAinG0VI8p.35b0
!
!
!
```

```
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport access vlan 10
!
interface FastEthernet0/2
 switchport access vlan 20
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
```

```
interface Vlan1
 no ip address
 shutdown
!
interface Vlan199
 ip address 192.168.99.4 255.255.255.192
!
ip default-gateway 192.168.99.1
!
!
!
!
line con 0
!
line vty 0 4
 password test..123
 login
line vty 5 15
 password test..123
 login
!
!
!
!
end
```

- SW_Server

```
Current configuration : 1399 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
enable secret 5 $1$mERr$oF4ioZ7MAinG0VI8p.35b0
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport access vlan 100
!
interface FastEthernet0/2
 switchport access vlan 100
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
 switchport access vlan 199
 switchport mode access
!
```

```
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
 switchport access vlan 99
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
 no ip address
 shutdown
!
interface Vlan199
 ip address 192.168.99.5 255.255.255.192
!
ip default-gateway 192.168.99.1
!
!
!
!
line con 0
!
line vty 0 4
 password test..123
 login
```

```
 line vty 5 15
  password test..123
  login
 !
 !
 !
 !
end
```

- DISTRI (L3 Switch)

```
Current configuration : 1666 bytes
!
version 16.3.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
enable secret 5 $1$mERr$oF4ioZ7MAinG0VI8p.35b0
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet1/0/1
 switchport mode trunk
!
interface GigabitEthernet1/0/2
 switchport mode trunk
!
interface GigabitEthernet1/0/3
 switchport mode trunk
!
interface GigabitEthernet1/0/4
```

```
!
interface GigabitEthernet1/0/5
!
interface GigabitEthernet1/0/6
!
interface GigabitEthernet1/0/7
!
interface GigabitEthernet1/0/8
!
interface GigabitEthernet1/0/9
!
interface GigabitEthernet1/0/10
!
interface GigabitEthernet1/0/11
!
interface GigabitEthernet1/0/12
!
interface GigabitEthernet1/0/13
!
interface GigabitEthernet1/0/14
!
interface GigabitEthernet1/0/15
!
interface GigabitEthernet1/0/16
!
interface GigabitEthernet1/0/17
!
interface GigabitEthernet1/0/18
!
interface GigabitEthernet1/0/19
!
interface GigabitEthernet1/0/20
!
interface GigabitEthernet1/0/21
!
interface GigabitEthernet1/0/22
!
interface GigabitEthernet1/0/23
!
interface GigabitEthernet1/0/24
 switchport mode trunk
!
interface GigabitEthernet1/1/1
!
interface GigabitEthernet1/1/2
!
interface GigabitEthernet1/1/3
!
interface GigabitEthernet1/1/4
!
interface Vlan1
 no ip address
 shutdown
!
interface Vlan199
 mac-address 00e0.a3ce.2a01
 ip address 192.168.99.2 255.255.255.192
!
ip default-gateway 192.168.99.1
ip classless
```

```
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
 password test..123
 login
line vty 5 15
 password test..123
 login
!
!
!
!
end
```

- EDGE ROUTER

```
Current configuration : 2554 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
enable secret 5 $1$mERr$oF4ioZ7MAinG0VI8p.35b0
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
```

```
!
!
!
!
!
interface GigabitEthernet0/0/0
 ip address 11.11.11.11 255.255.255.248
 ip nat outside
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/1
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/1.1
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
 ip helper-address 192.168.100.12
 ip nat inside
!
interface GigabitEthernet0/0/1.2
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
 ip helper-address 192.168.100.12
 ip access-group 101 in
 ip nat inside
!
interface GigabitEthernet0/0/1.3
 encapsulation dot1Q 50
 ip address 192.168.50.1 255.255.254.0
 ip nat inside
!
interface GigabitEthernet0/0/1.4
 encapsulation dot1Q 99
 ip address 192.168.99.97 255.255.255.240
 ip nat inside
!
interface GigabitEthernet0/0/1.5
 encapsulation dot1Q 100
 ip address 192.168.100.1 255.255.255.0
 ip nat inside
!
interface GigabitEthernet0/0/1.6
 encapsulation dot1Q 199
 ip address 192.168.99.1 255.255.255.192
 ip nat inside
!
interface GigabitEthernet0/0/1.7
 encapsulation dot1Q 299
 ip address 192.168.99.65 255.255.255.224
 ip nat inside
!
interface Vlan1
 no ip address
 shutdown
!
router rip
!
```

```
ip nat pool NAT-POOL 11.11.11.11 11.11.11.11 netmask 255.255.255.248
ip nat inside source list 10 interface GigabitEthernet0/0/0 overload
ip nat inside source static tcp 192.168.99.98 80 11.11.11.11 80
ip nat inside source static tcp 192.168.99.98 443 11.11.11.11 443
ip classless
ip route 8.8.8.0 255.255.255.0 11.11.11.10
ip route 78.79.80.0 255.255.255.0 11.11.11.10
ip route 104.244.42.0 255.255.255.0 11.11.11.10
ip route 158.64.75.0 255.255.255.0 11.11.11.10
ip route 22.22.22.16 255.255.255.248 11.11.11.10
ip route 33.33.33.32 255.255.255.248 11.11.11.10
!
ip flow-export version 9
!
!
access-list 10 permit 192.168.10.0 0.0.0.255
access-list 10 permit 192.168.20.0 0.0.0.255
access-list 10 permit 192.168.50.0 0.0.1.255
access-list 10 permit 192.168.99.96 0.0.0.15
access-list 10 permit 192.168.100.0 0.0.0.255
access-list 10 permit 192.168.99.0 0.0.0.63
access-list 10 permit 192.168.99.64 0.0.0.31
access-list 101 deny ip any host 192.168.99.98
access-list 101 permit ip any any
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
 password test..123
 login
line vty 5 15
 password test..123
 login
!
!
!
end
```

- ISP Router

```
Current configuration : 908 bytes
!
version 12.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
!
```

```
!
no ip cef
no ipv6 cef
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
interface GigabitEthernet0/0
 ip address 22.22.22.21 255.255.255.248
 duplex auto
 speed auto
!
interface GigabitEthernet1/0
 ip address 33.33.33.34 255.255.255.248
 duplex auto
 speed auto
!
interface GigabitEthernet2/0
 ip address 1.1.1.101 255.255.255.252
 duplex auto
 speed auto
!
interface GigabitEthernet3/0
 ip address 11.11.11.10 255.255.255.248
 duplex auto
 speed auto
!
router rip
!
ip classless
ip route 8.8.8.0 255.255.255.0 1.1.1.102
ip route 158.64.75.0 255.255.255.0 1.1.1.102
ip route 78.79.80.0 255.255.255.0 1.1.1.102
ip route 104.244.42.0 255.255.255.0 1.1.1.102
!
ip flow-export version 9
!
!
!
!
!
!
!
!
line con 0
!
```

```
 line aux 0
 !
 line vty 0 4
  login
 !
 !
 !
end
```

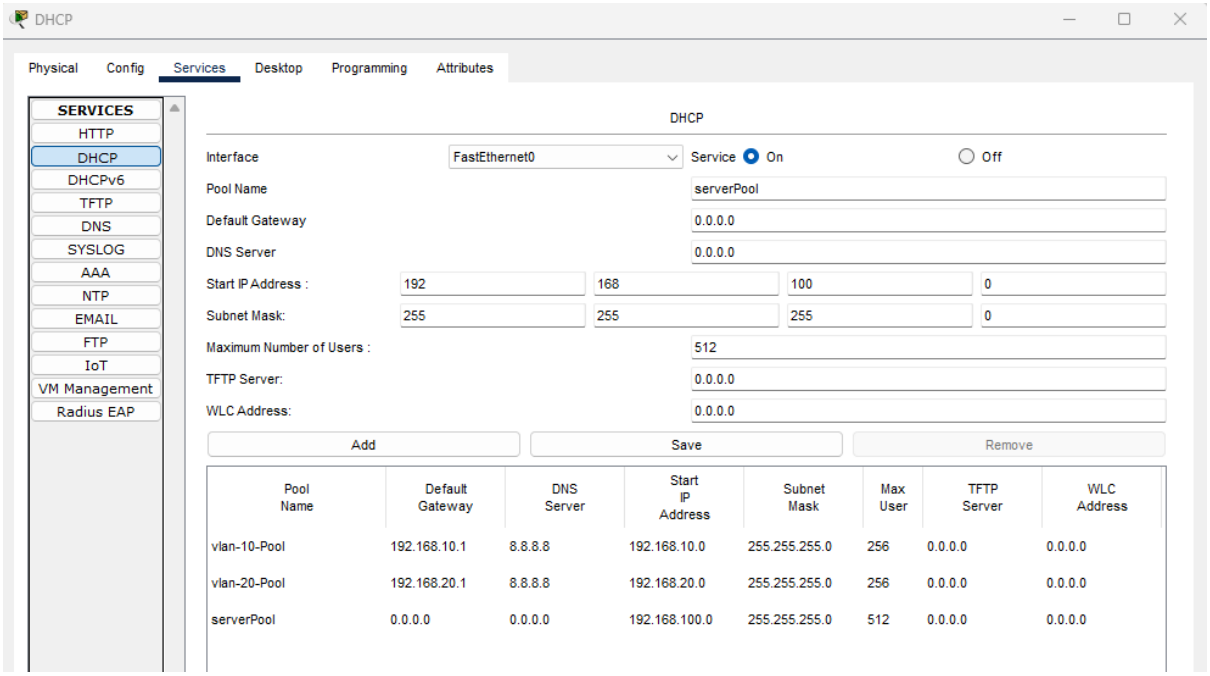
- INET Router

```
 Current configuration : 955 bytes
 !
 version 12.2
 no service timestamps log datetime msec
 no service timestamps debug datetime msec
 no service password-encryption
 !
 hostname Router
 !
 !
 !
 !
 !
 !
 !
 !
 no ip cef
 no ipv6 cef
 !
 !
 !
 !
 !
 !
 !
 !
 !
 !
 !
 !
 !
 !
 !
 !
 !
 !
 !
 interface GigabitEthernet0/0
  ip address 1.1.1.102 255.255.255.252
  duplex auto
  speed auto
 !
 interface GigabitEthernet1/0
  ip address 8.8.8.1 255.255.255.0
  duplex auto
  speed auto
 !
 interface GigabitEthernet2/0
  ip address 158.64.75.1 255.255.255.0
  duplex auto
  speed auto
```

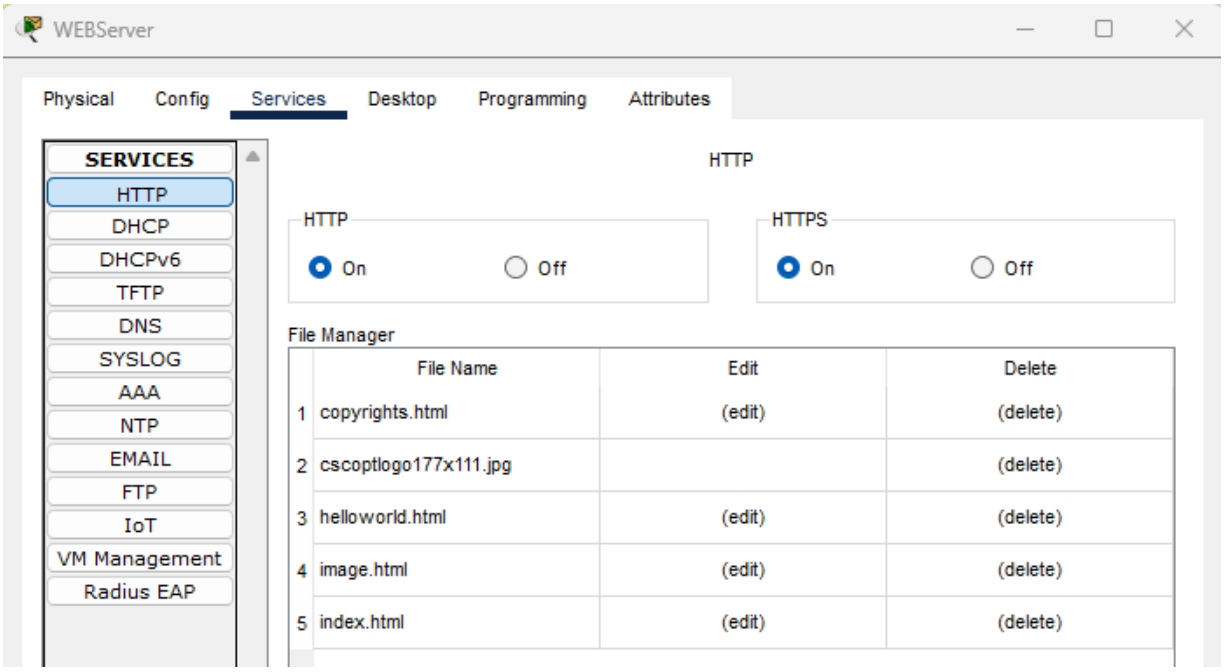
```
!
interface GigabitEthernet3/0
 ip address 78.79.80.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet4/0
 ip address 104.244.42.1 255.255.255.0
 duplex auto
 speed auto
!
router rip
!
ip classless
ip route 11.11.11.8 255.255.255.248 1.1.1.101
ip route 22.22.22.16 255.255.255.248 1.1.1.101
ip route 33.33.33.32 255.255.255.248 1.1.1.101
!
ip flow-export version 9
!
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
 login
!
!
!
end
```

Configuration des Services des Serveurs

DHCP

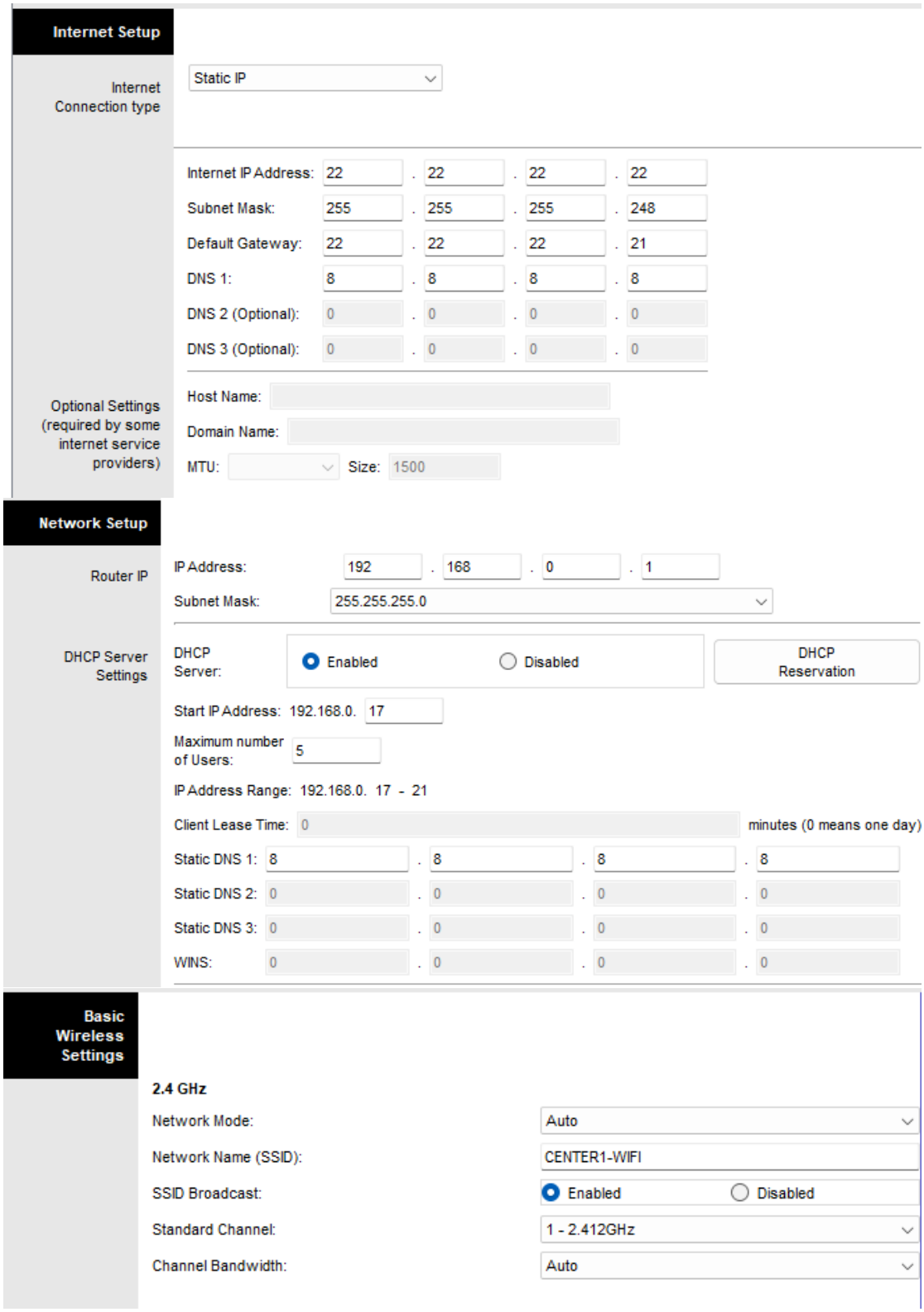


WEB



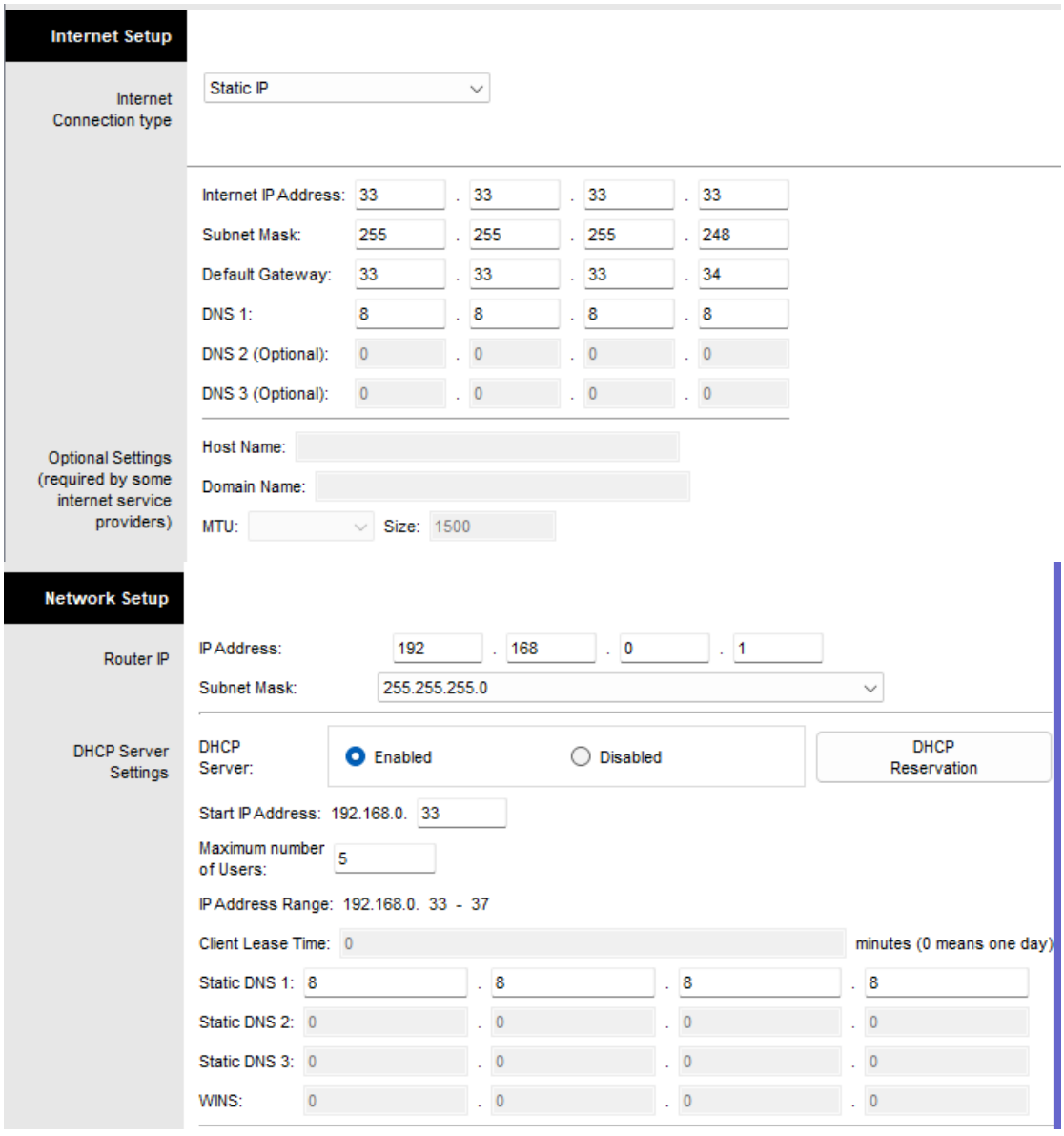
Configuration des HOME-Routers

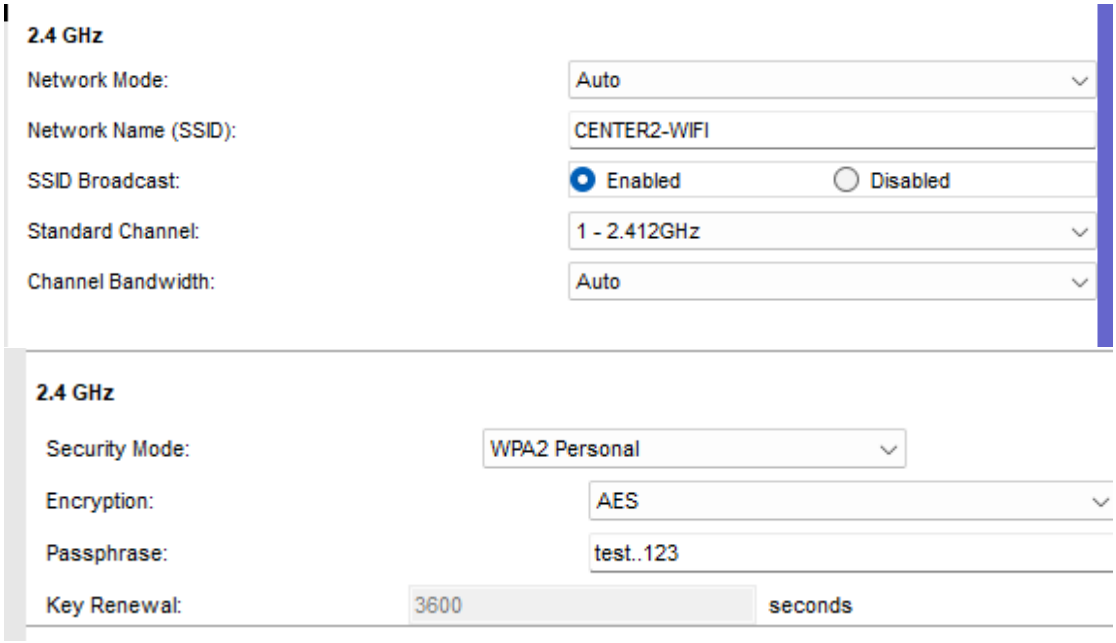
CENTER1




Wireless Security
2.4 GHz
Security Mode:
WPA2 Personal
Encryption:
AES
Passphrase:
test..123
Key Renewal:
3600
seconds

CENTER2

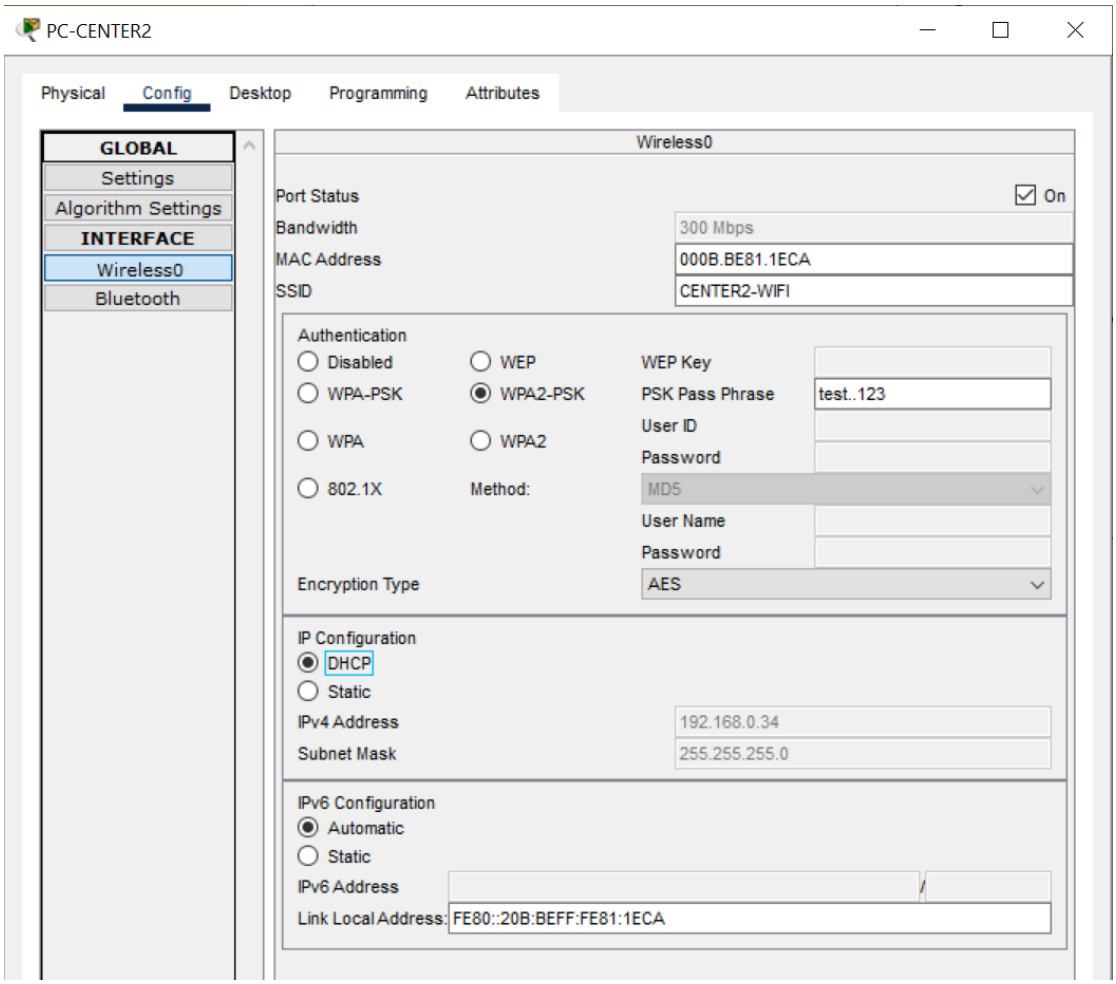

Internet Setup
Internet Connection type
Static IP
Internet IP Address: 33 . 33 . 33 . 33
Subnet Mask: 255 . 255 . 255 . 248
Default Gateway: 33 . 33 . 33 . 34
DNS 1: 8 . 8 . 8 . 8
DNS 2 (Optional): 0 . 0 . 0 . 0
DNS 3 (Optional): 0 . 0 . 0 . 0
Optional Settings (required by some internet service providers)
Host Name:
Domain Name:
MTU:
Size: 1500
Network Setup
Router IP
IP Address: 192 . 168 . 0 . 1
Subnet Mask: 255.255.255.0
DHCP Server Settings
DHCP Server:
Enabled
Disabled
DHCP Reservation
Start IP Address: 192.168.0. 33
Maximum number of Users: 5
IP Address Range: 192.168.0. 33 - 37
Client Lease Time: 0
minutes (0 means one day)
Static DNS 1: 8 . 8 . 8 . 8
Static DNS 2: 0 . 0 . 0 . 0
Static DNS 3: 0 . 0 . 0 . 0
WINS: 0 . 0 . 0 . 0


2.4 GHz
Network Mode:
Auto
Network Name (SSID):
CENTER2-WIFI
SSID Broadcast:
Enabled
Disabled
Standard Channel:
1 - 2.412GHz
Channel Bandwidth:
Auto

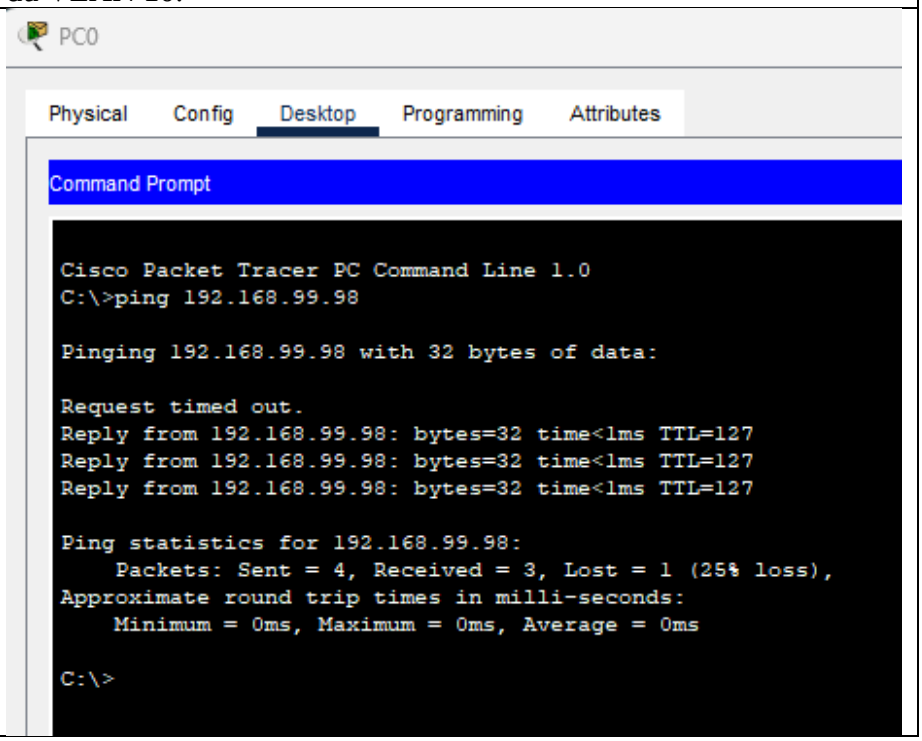
2.4 GHz
Security Mode:
WPA2 Personal
Encryption:
AES
Passphrase:
test..123
Key Renewal:
3600
seconds

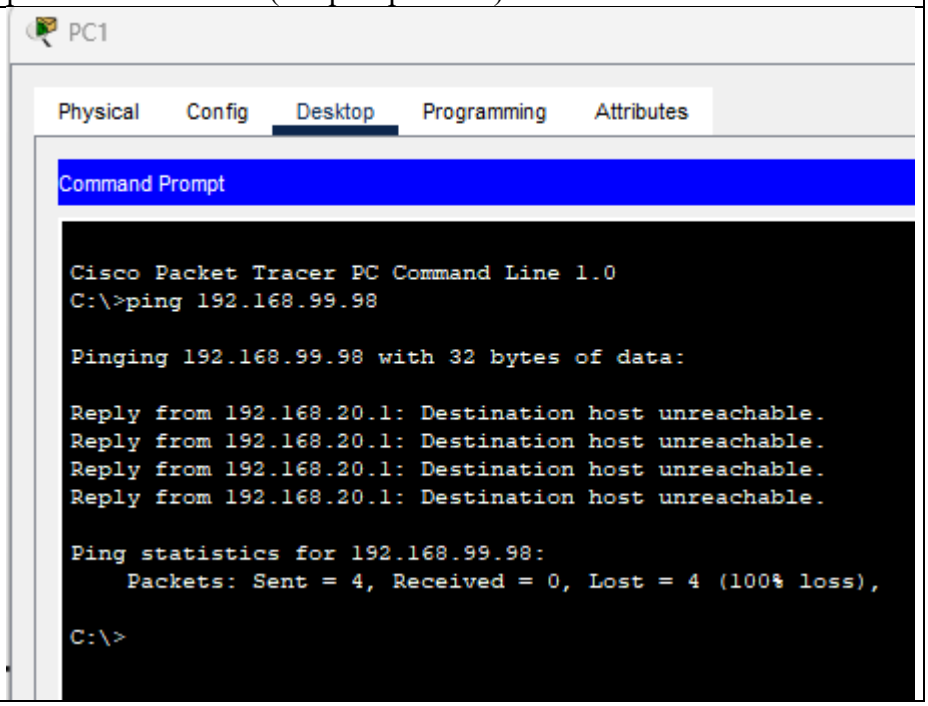
Configuration des appareils sans fils

PC CENTER2

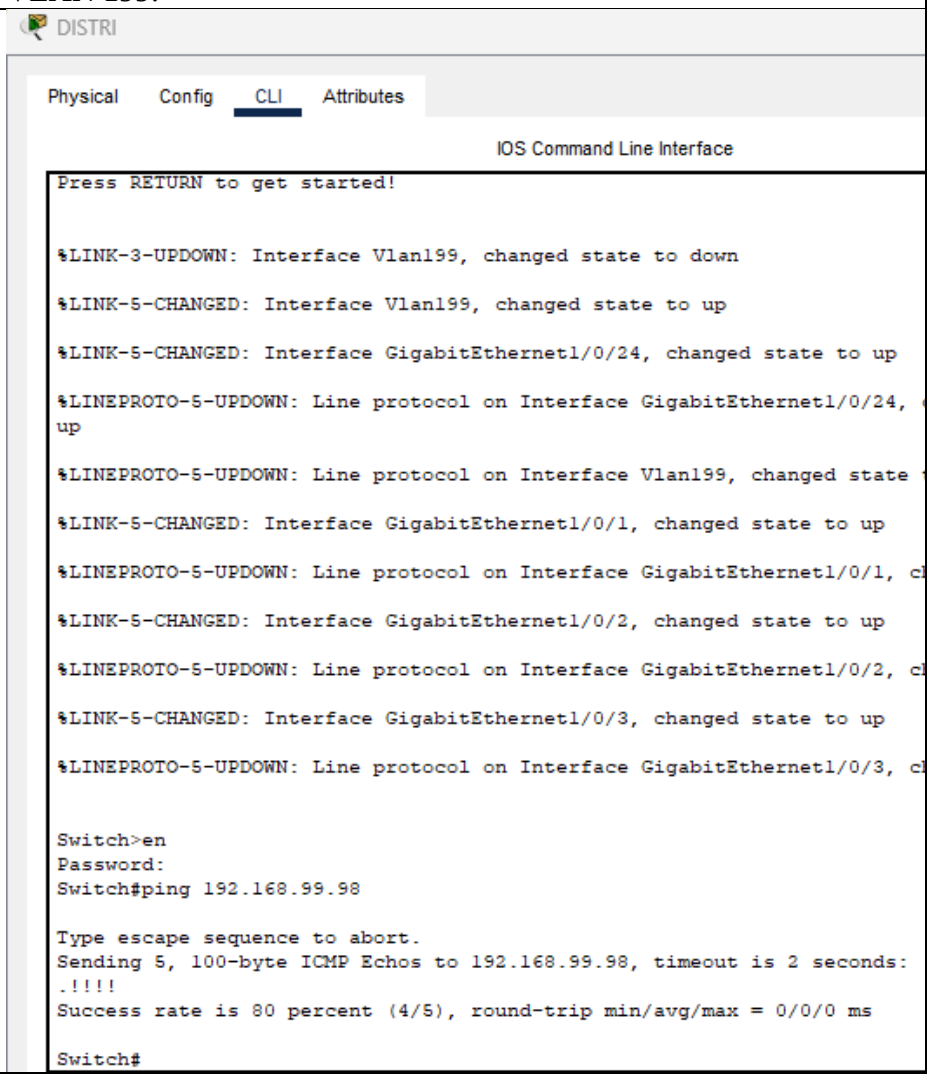


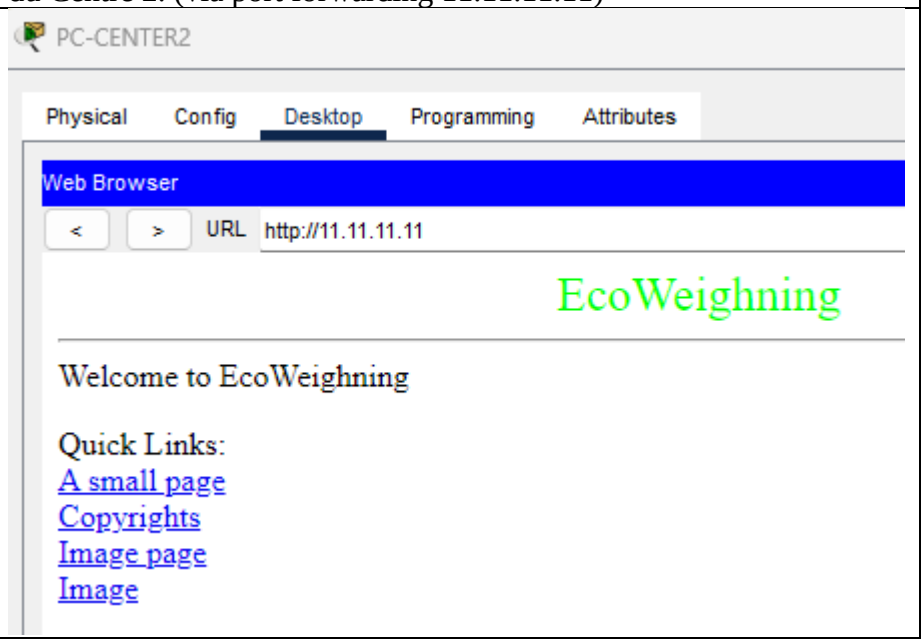
Protocol de Vérification (Prüfprotokolle)

Test 1 : Ping du PC0 au Serveur Web/Base de données	
Date	30, Janvier 2024
Source	PC0 (VLAN 10 / BENUTZER 1)
Destination	Serveur Web/Base de données
But du test	Montre que le Serveur Web/Base de données est accessible à partir du VLAN 10.
Screenshot	

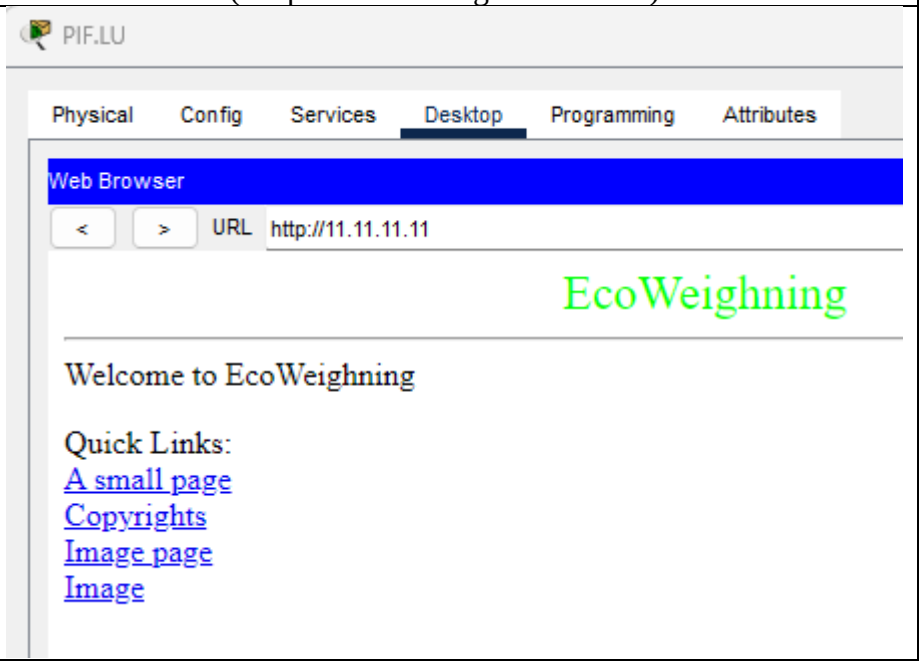
Test 2 : Ping du PC1 au Serveur Web/Base de données	
Date	30, Janvier 2024
Source	PC1 (VLAN 20 / BENUTZER 2)
Destination	Serveur Web/Base de données
But du test	Montre que le Serveur Web/Base de données n'est pas accessible à partir du VLAN 20. (bloquée par l'acl)
Screenshot	

Test 3 : Requête HTTP du Serveur Intranet au Serveur Web/Base de données	
Date	30, Janvier 2024
Source	Serveur Intranet (VLAN 100 / SERVER)
Destination	Serveur Web/Base de données
But du test	Montre que le Serveur Web/Base de données est accessible à partir du VLAN 100.
Screenshot	

Test 4 : Ping du L3 Switch (DISTRI) au Serveur Web/Base de données	
Date	30, Janvier 2024
Source	L3 Switch / DISTRI (VLAN 199 / MGMT)
Destination	Serveur Web/Base de données
But du test	Montre que le Serveur Web/Base de données est accessible à partir du VLAN 199.
Screenshot	

Test 5 : Requête HTTP du PC CENTER 2 au Serveur Web/Base de données	
Date	30, Janvier 2024
Source	PC CENTER 2 (réseaux du Centre 2)
Destination	Serveur Web/Base de données
But du test	Montre que le Serveur Web/Base de données est accessible à partir du Centre 2. (via port forwarding 11.11.11.11)
Screenshot	

Test 6 : Requête HTTP du PC CENTER 1 au Serveur Web/Base de données	
Date	30, Janvier 2024
Source	PC CENTER 1 (réseaux du Centre 1)
Destination	Serveur Web/Base de données
But du test	Montre que le Serveur Web/Base de données est accessible à partir du Centre 1. (via port forwarding 11.11.11.11)
Screenshot	

Test 7 : Requête HTTP du Serveur pif.lu au Serveur Web/Base de données	
Date	30, Janvier 2024
Source	pif.lu (internet)
Destination	Serveur Web/Base de données
But du test	Montre que le Serveur Web/Base de données est accessible à partir du Web/internet. (via port forwarding 11.11.11.11)
Screenshot	

Fonctionnalités Optionnelles

- Ajout d'un pc portable pour configurer les routeurs et les switches via le vlan 199. (également ajouté au fichier du must)
- Le Serveur DNS de google est fonctionnel et est utilisé par les appareils en dehors du réseau interne d'ecoweighing.
- Ajout d'un Serveur DNS interne au réseau d'ecoweighing et est utilisé par tous les appareils interne.
- Les Serveurs google.com, education.lu, pif.lu et twitter.com ont une page web accessible par tous les appareilles avec leur propre nom de domaine (DNS).
- (! Pas Terminer !) Wlan du réseaux ecoweighing avec WLC.